MAST30027: Modern Applied Statistics

Exam Solutions

1. **Question 5**

   (a) Show that the Bayes estimator under the given loss function is $\hat{\theta} = (E[\theta^{-1}|\mathbf{x}])^{-1}$.

   Solution: $E_\theta[R_{\hat{\theta}}(\theta)] = E_\theta[E_{\mathbf{X}}[\frac{(\hat{\theta}-\theta)^2}{\theta}|\theta]] = E_{\mathbf{X}}(E_\theta[\frac{(\hat{\theta}-\theta)^2}{\theta}|\mathbf{X}])$. Hence for each given sample $\mathbf{x}$, the Bayes estimator $\hat{\theta} = \arg\min_\theta E_\theta[\frac{(\hat{\theta}-\theta)^2}{\theta}|\mathbf{x}]$. Now $E_\theta[\frac{(\hat{\theta}-\theta)^2}{\theta}|\mathbf{x}] = E[\theta^{-1}|\mathbf{x}]\hat{\theta}^2 - 2\hat{\theta} + E[\theta|\mathbf{x}]$ which is a quadratic function of $\hat{\theta}$. It is easy to see that it is minimised when $\hat{\theta} = (E[\theta^{-1}|\mathbf{x}])^{-1}$ which gives the Bayes estimator of θ.